kotlin

Kotlin syntax vs JavaScript/TypeScript

Concept	JavaScript/TypeScript	Kotlin
Variable	`let name = "John"`	`var name = "John"`
Constant	`const name = "John"`	`val name = "John"`
Function	`function hello() {}`	`fun hello() {}`
Type	`let age: number = 5`	`var age: Int = 5`
Null safe	`name?: string`	`var name: String?`
Class	`class Foo extends Bar`	`class Foo : Bar()`
Print	`console.log("hi")`	`println("hi")`
Arrow fn	`const f = () => {}`	`val f = {}`

Class oop

```
//* properties (Under class name)
private lateinit var value:TextView
private var counter: Int = 0

//* init (Under setContentView)
value = findById<TextView>(R.id.value)

//* Method (Under setContentView)
m1()
m2()
```

Methods

```
EditTextElement.getText()
TextViewElement.toString()
```

Toast

```
Toast.makeText(this, "text here", Toast.LENGTH_SHORT).show()
```

Data Binding

is Simple method to call element without using `findById`

1. Enable `dataBinding`

Go `build.grade (Module)` add this:

```
buildFeatures{
    dataBinding = true
}
```

then click on `Sync now`

2. wrap your xml content with layout:

```
<layout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto">
// content here
</layout>
```

3. Add these to kotlin file:

```
private lateinit var binding: ActivityMainBinding //* name of xml + Binding

//* change setContent(..) BY this
binding = DataBindingUtil.setContentView(this, R.layout.activity_main)

//* then use it like
binding.elementId
```

Intent

1. phone

```
val phone = binding.phoneId

phone.setOnClickLinstener {

    val intent = Intent(Intent.ACTION_DIAL).apply {
        data = Uri.parse("tel:${phone.text.toString()})
    }

    intent.startActivity()
}
```

2. email

```
val email = binding.email
email.setOnClickListener {
    val intent = Intent(Intent.ACTION_SENDTO).apply{
        data = Uri.parse("mailto:${email.text.toString()}")
    }
    startActivity(intent)
}
```

3. website

```
val website = binding.website
website.setOnClickListener {
    val intent = Intent(Intent.ACTION_VIEW).apply{
        data = Uri.parse(website.text.toString())
    }
    startActivity(intent)
}
```

4. map address

```
val map = binding.address
map.setOnClickListener {
    val addressEncoded = Uri.encode(map.text.toString())
    val intent = Intent(Intent.ACTION_VIEW).apply{
        data = Uri.parse("geo:0,0?q=${addressEncoded}")
    }
    startActivity(intent)
}
```

5. Camera

```
profilePhoto = binding.profilePhoto
profilePhoto.setOnClickListener {
    val intent = Intent(MediaStore.ACTION_IMAGE_CAPTURE)
    cameraLauncher.launch(intent)
}

private val cameraLauncher = registerForActivityResult(
    ActivityResultContracts.StartActivityForResult()
) { result ->
    if (result.resultCode == RESULT_OK) {
        // get the thumbnail photo from camera
        val photo = result.data?.extras?.get("data") as?
android.graphics.Bitmap
        photo?.let { profilePhoto.setImageBitmap(it) }
```

```
    }
}
```